

URBAN MOBILITY OPTIMIZATION SYSTEM (UMOS)

A Multimodal Routing Platform for University Transport in Bogotá

PROBLEM



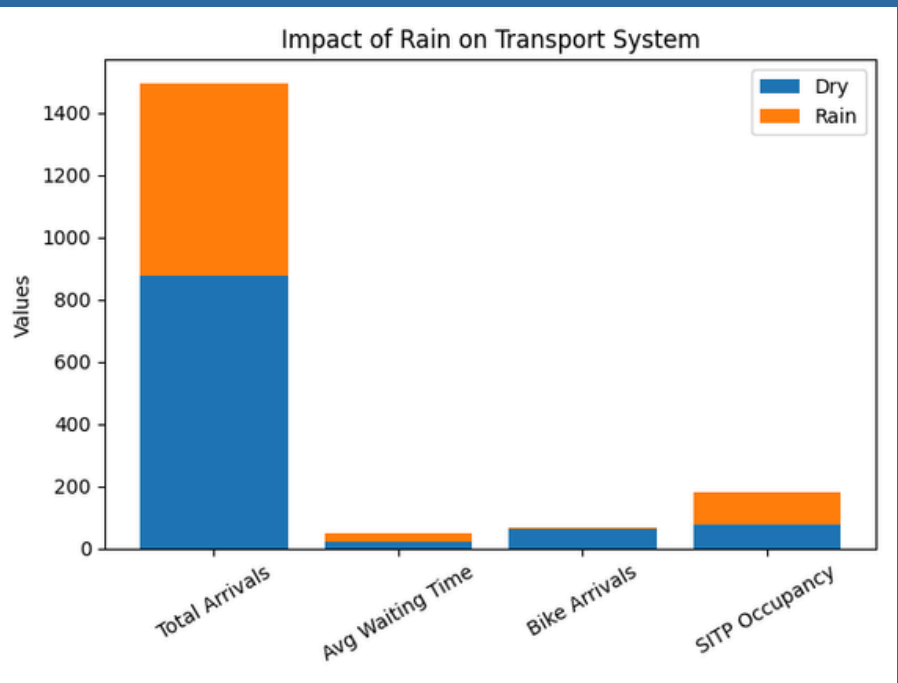
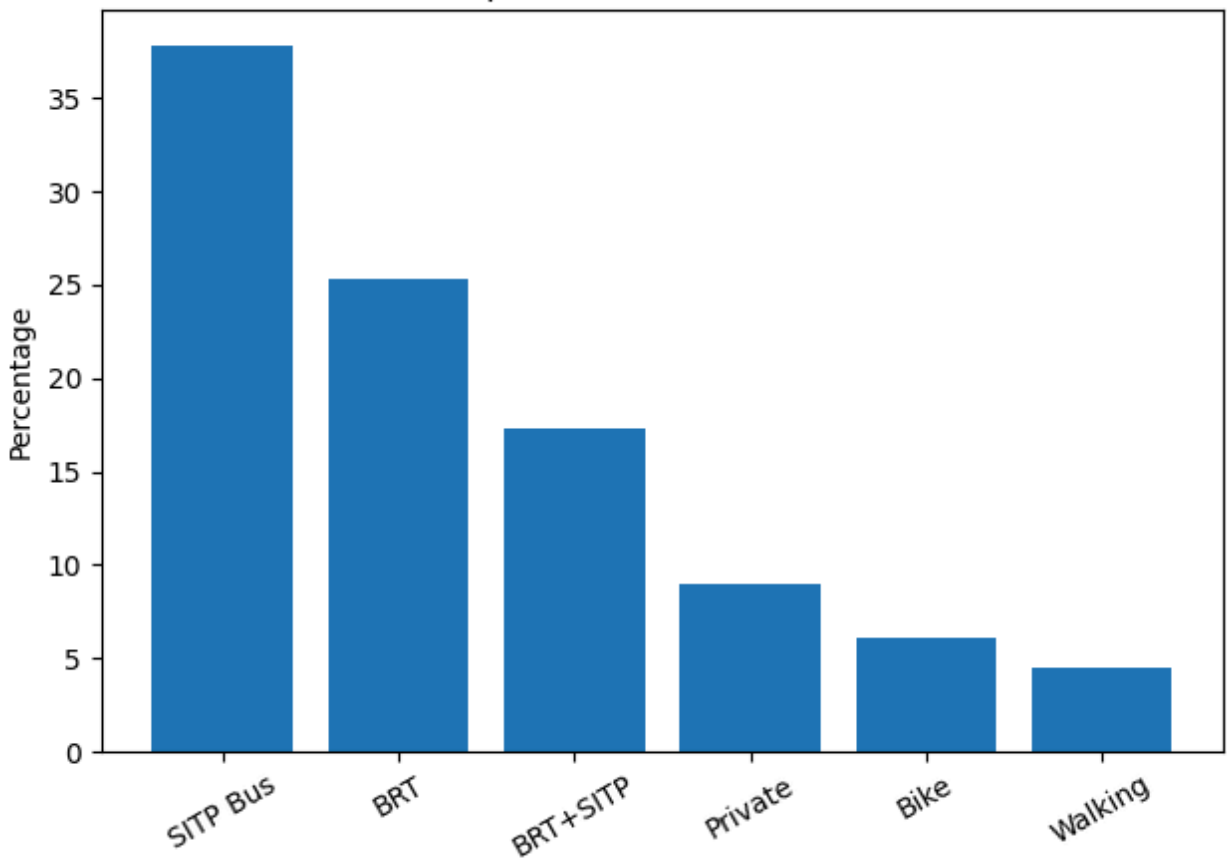
- Public transportation in Bogotá is highly unpredictable and inefficient.
- Average commute: 67 minutes per trip
- Up to 82 minutes with multimodal transport
- Rain increases: Waiting time by 63%
- System variability by 128%
- Lack of real-time, reliable information for users

PROPOSED SOLUTION



- UMOS is a mobile application that provides:
- Real-time multimodal route recommendations
- Integration of:
 - BRT (TransMilenio)
 - SITP buses
 - Cable transport
 - Cycling and walking routes
- Dynamic adaptation to:
 - Traffic conditions
 - Weather changes
 - User reports

Transport Mode Distribution (%)



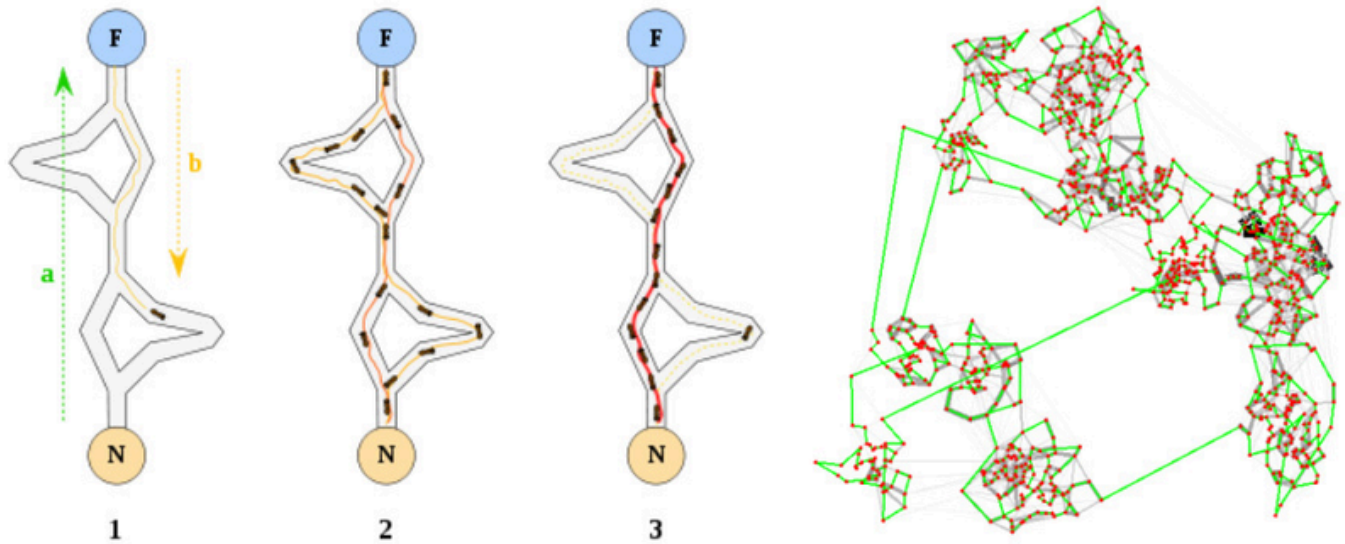
SYSTEM ARCHITECTURE

- Microservices-based system designed for scalability and real-time processing
- Composed of modules for routing, data integration, notifications, and analytics
- Communication via REST, gRPC, and event streams
- Handles up to 14,000 GPS events per minute



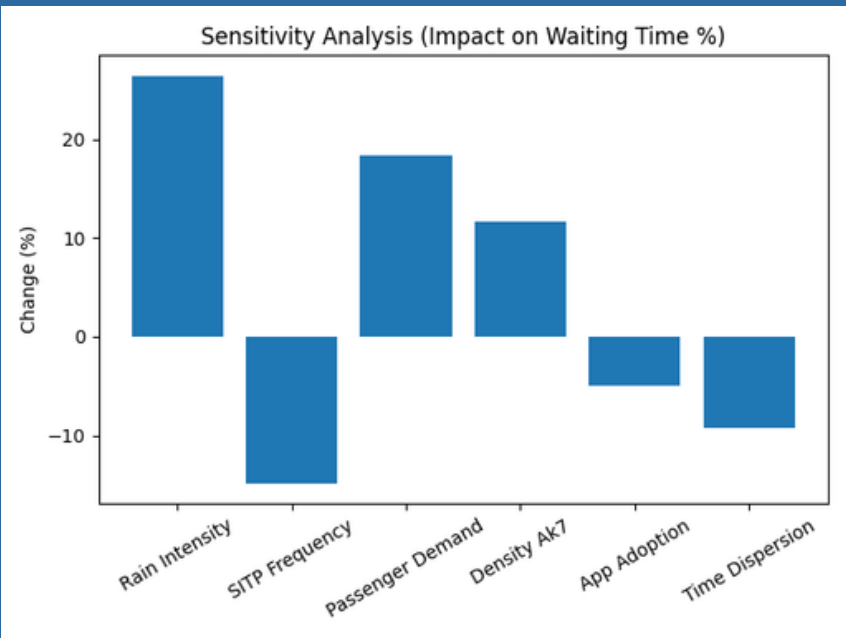
ROUTING APPROACH

- Traditional algorithms fail under dynamic conditions
- UMOS uses Ant Colony Optimization (ACO)
- Key features:
 - Adapts to real-time traffic and weather
 - Learns from previous route performance
 - Generates optimal routes in < 800 ms



DATA & INSIGHTS

- Based on 312 surveys + field observations
- Findings:
 - Public transport dominates usage
 - Rain increases waiting time (+63%)
 - High variability makes the system unpredictable



PERFORMANCE



- Response time: < 2 seconds
- Availability: 99.5%
- Uses caching and event-driven updates
- Supports real-time decision making

IMPACT

- Reduces travel uncertainty
- Improves route selection
- Enhances user experience
- Provides real-time mobility insights

